Scheduling in Higher Education: Purdue University as a Case Study

Samantha Stutler · sstutler@purdue.edu · AAE, Engineering · John Martinson Honors College, Purdue University

Faculty mentor: Professor Davi Moreira, Mitch Daniels School of Business

HEADLINE **College of Health and Human Sciences at Purdue West Lafayette in 2026 has significantly greater proportion of classes before 9 AM compared to nine other colleges only when all instructional formats are considered. Lecture sections do not follow the same pattern.**

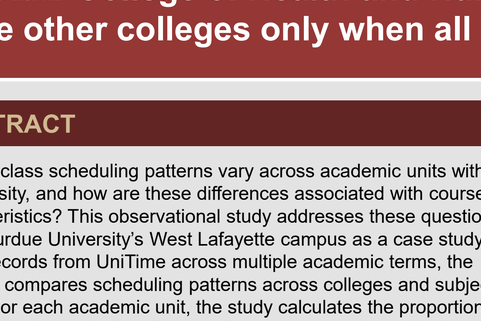
ABSTRACT

How do class scheduling patterns vary across academic units within a university, and how are these differences associated with course characteristics? This observational study addresses these questions using Purdue University's West Lafayette campus as a case study. Using records from UniTime across multiple academic terms, the analysis compares scheduling patterns across colleges and subject areas. For each academic unit, the study calculates the proportions of sections beginning before 9 a.m., after 5 p.m., together with the mean start time. Additional comparisons examine how these patterns vary by course level, meeting duration, enrollment, section capacity, and instructional format, while comparisons across terms assess the consistency of observed patterns. The study characterizes the distribution of class times and identifies patterns that warrant further investigation. The case study provides an empirical foundation for discussions of scheduling equity and resource allocation, while offering a replicable framework for comparative research across higher education institutions.

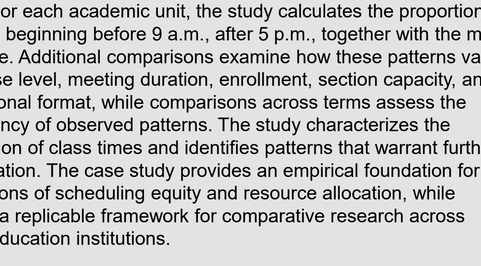
THE QUESTION AND WHY IT MATTERS

How do class scheduling patterns vary across academic units within a university, and how are these differences associated with course characteristics? This matters to any student or professor beholden to academic scheduling, as scheduling has been shown to have significant effects and associations on outcomes (grades, program success).

F26 NUR Schedule



F26 AAE Schedule



WHAT IS ALREADY KNOWN

The main conclusion from the literature is that different scheduling patterns– early class times (Yeo et al, 2023), back to back classes (Williams & Shapiro, 2018) and the "power hour" structure (Botswick & Nguyen, 2026) –are associated with different student outcomes. This project adds data analysis on the ways different majors are scheduled and scheduling outcomes at Purdue.

DESIGN: WHAT YOUR CLAIM NEEDS

KIND	Descriptive, non-causal.
REACH	Data at hand for Purdue WL within 2026 time frame, population for any other application. Unseen cases include classes not scheduled in UniTime.
TARGET	Four metrics: proportion of classes before 9am, after 5pm, outside of the 9 to 5, and average class start time
WARRANT	This only holds for the population of Fall and Spring 2026 for Purdue West Lafayette, which was the data measured

DATA AND MEASUREMENT

Source and route	UniTime Scheduling system, F26 retrieved on 9/26, S26 retrieved on 10/4
Observations	32518 rows and 165 sections in S26, 38350 rows and 167 sections in F26. 12 colleges.
Independent units	Unique sections
Time window	Fall and Spring 2026, 2024 and 2025 data to be added soon
Excluded data	Sections with time "Arrange hours", type "Travel Time" or zero enrollment
Permissions	IDP gave permission for UniTime data, IRB gave permission for reviews

METHODS

- MATLAB flowchart:



- Confidence levels: z = 5 for subjects, 99.9% for colleges; the level required to bring false positives to acceptable levels
- Counting rules: before 9 AM, after 5 PM, outside 9-to-5, sections with multiple times (for proportions and average), sections with 2+ subjects or colleges, what is a college

MAIN RESULT

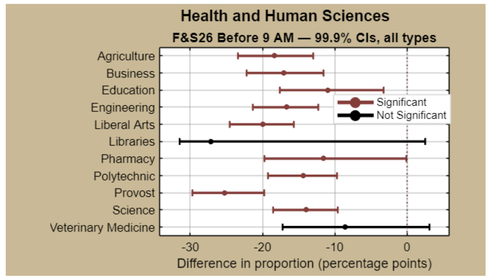


Fig. 1. All 99.9% difference interval comparisons for before 9 AM proportion for the College of Health and Human Sciences for all instructional formats. Produced with MATLAB with data from UniTime.

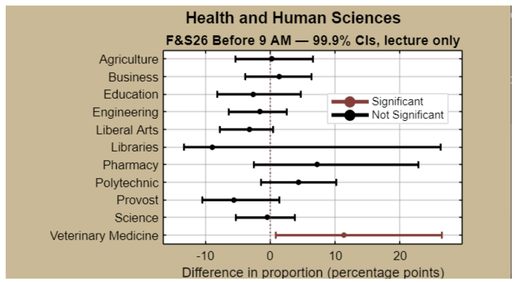


Fig. 2. All 99.9% difference interval comparisons for before 9 AM proportion for the College of Health and Human Sciences for lecture sections only. Produced with MATLAB with data from UniTime.

The number of colleges that HHS's before 9 Am proportion is significantly different from drops from 9 to 1 when non-lecture courses are filtered out. The 99.9% interval filters out that the difference could be due to "chance" in scheduling, indicating correlation between average start time, college, and instructional format.

CHECKS AND VERIFICATION

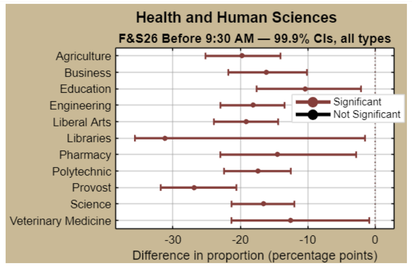


Fig. 3. All 99.9% difference interval comparisons for before 9:30 AM proportion for the College of Health and Human Sciences for all instructional formats. Produced with MATLAB with data from UniTime. This check shows how much results change when arbitrary cutoff changes.

CONCLUSIONS

In Fall and Spring 2026, on Purdue's West Lafayette campus, classes scheduled through UniTime have trends with the average start times, proportion outside of the 9-to-5, and other metrics on the subject and college level. There are additional trends for the instructional format and other confounding factors. Because of the literature showing that early classes lead to negative outcomes, we can conclude that students would be more likely to struggle in HHS classes.

LIMITATIONS

Uncertainty: between S26 and F26, the % difference in average start time for each subject is 4.51%. 2.19% for subjects with 20+ sections.

Limitation: only used 2 semesters of data. Patterns are not explained.

Next Steps

Analyze the 2024 and 2025 data that is already obtained. Conduct interviews for explanation. [check: Conduct Anova test] Use a regression to check associations with other variables. Observe trends around the change in scheduling around [demolished building's] closing.

AI USE DISCLOSURE

For this project, ChatGPT, Copilot and Gemini were used. These tools were used for ideation, as a search engine, and for coding assistance. I take responsibility for all conclusions.

REFERENCES

Yeo, Sing Chen, et al. "Early morning university classes are associated with impaired sleep and academic performance." Nature Human Behaviour 7.4 (2023): 502-514. https://www.nature.com/articles/s41562-023-01531-x

Williams, Kevin M., and Teny Maghakian Shapiro. "Academic achievement across the day: Evidence from randomized class schedules." Economics of Education Review 67 (2018): 158-170. https://www.sciencedirect.com/science/article/pii/S0272775717305150

Bostwick, Valerie, and Tuan Nguyen. "More often or longer? The effects of the academic schedule on postsecondary academic outcomes." Economics of Education Review 113 (2026): 102812. https://www.sciencedirect.com/science/article/pii/S0272775726000543

ACKNOWLEDGMENTS

I thank Professor Davi Moreira for mentoring this project, and the John Martinson Honors College and the Office of Undergraduate Research for the chance to present it. Thank you to UniTime and Internal Data processing for providing data.